

10..9..8..

Game Design Document

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Executive Summary

Elevator pitch

10..9..8.. is a dungeon exploration game in which the player can only move 10 steps.

Summary

10..9..8.. is a 2D top-view with a fast-paced roguelike game, inspired by chess.

Unlike traditional chess, this game has unique enemies that threaten the player's journey

To the final boss, the chess king, can you survive in the dungeon and checkmate the King?

Key Features

- A variety of items for dungeon exploration
- Different concepts for every stage
- Well-matched music tracks for every stage
- Chess-style design

Target Audience

- Teens and adults ages 13+
- Chess fans
- Roguelike fans

Target Platform

- Microsoft Windows 10 or the latest
- 1024*768 fixed resolution

Story

Prologue

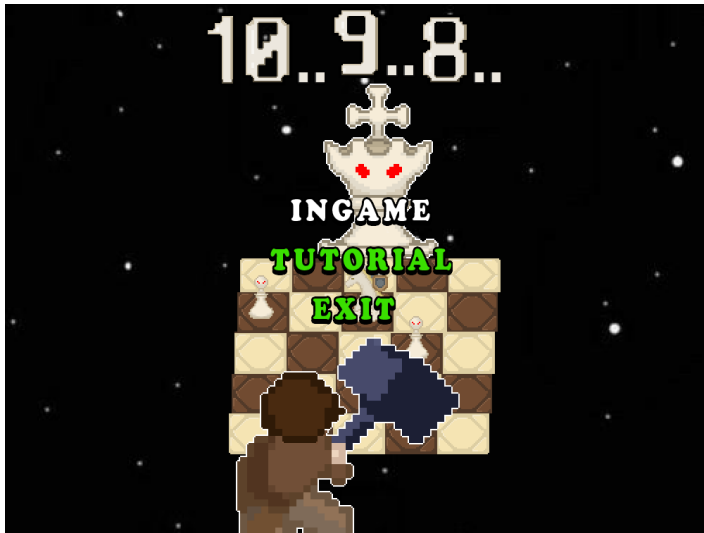
The player embarks on a journey to confront a cursed kingdom ruled by the evil Chess King.
The goal is to destroy all the living chess statues and defeat the evil Chess King.

Game Flow

Splash Screens

When the game starts, short splash screens of the Digipen logo.

Title Screen



The title screen shows a large picture of the player with Chess King encountered in the background. From the title screen, players will choose from two game modes, In-game and Tutorial.

In Game

In the In game, players begin from stage 1-1 with a randomly selected map. Players move through the dungeon in 10 steps while fighting or avoiding enemies. If the player successfully reaches the stairs, players can advance to the next floor. Each stage consists of 3 floors, and the 4th floor is at a store level. In the In game, players begin from stage 1-1 with a randomly selected map. Players move through the dungeon in 10 steps. Once the player leaves the store, the next stage is loaded.

Tutorial

In this mode, Players can learn how each enemy moves. From the basic enemy pawn to the long-ranged enemy, players will be familiar with enemy behavior. After observing every type of enemy, players will also visit the store level, where they can practice purchasing items using the remaining steps.

Game Mechanics

Turn System

This game is a turn-based roguelike game, so it is due to manage the turns for each enemy and player Turn manager is necessary.

- If the player, attacked by the enemy, the Sub(amount) function is used to decrease the remaining steps. This function is also used when taking steps to buy items in the store. This function always decreases the steps every time the player moves.
- If the player eats the item, restore the player's steps by the Add(amount) function.

Show Collision

Similar to how CS230 displays a collision box, this feature shows the enemy's next movement when the player toggles the 'Tab' key. Since the game has unique behavior of enemies, unlike chess, players would not know the movement of the enemy, so this shows the enemy's possible movement by coloring the tile red.

Map

To make the game more diverse, this game contains a semi-random map generator.

First, it opens the stage folder and scans through the names. There are 3 different text files for one single map,

1. A file with a single number name(ex. 1.txt) contains the tile's layout data.
2. A file with a single number name followed by _m (ex., 1_m.txt) contains information about enemy types and their spawn locations on the tile.
3. A file with a single number followed by _t (ex., 1_t.txt) contains information about the locations where traps are placed.

Second, random numbers are generated using the compiler, and the order of the numbers is shuffled.

Lastly, generate the map layout using the first three numbers from the shuffled list.

After that, read the tile numbers and place the corresponding tiles accordingly.

Game Mechanics - continued

Item Manager

This game features a variety of random items.

First, when a player defeats an enemy, the enemy drops an item that restores the player's steps.

Second, players can purchase randomly generated items in the store.

To manage this randomness, an Item Manager is required. Items are determined based on predefined probability rules. Each item has a base probability of appearing, which is influenced by its item class.

There is a dedicated logic system for handling the player's weapons. Players can purchase two different, distinct types of weapons, each with its durability. These weapons possess powerful abilities, which will be detailed later. This system is responsible for calculating each weapon's range and unique effects.

How To Win

To complete an individual level, the player must reach the stairs in the dungeon. Each stage contains stairs that lead to the next floor. And the final floor of every stage is always a store level. If the player reaches the store on Stage 3(i.e., 3-4), the next stage will be a boss level.

If the player defeats the boss, they win the game.

How To Lose

If the player loses all their steps, whether by being attacked by an enemy, stepping into an activated trap, purchasing an unaffordable item, or simply overusing steps, the game will end immediately.

Game Characters



Player

The player is the character controlled by the player and uses a massive hammer to destroy enemies with powerful strikes.



Pawn

Pawn is the basic enemy. It moves one tile per turn in the four cardinal directions(up, down, left, or right) and attacks the player if within range. When it attacks, it reduces the player's steps by 1.



Bishop

Bishop is a multi-tile mover enemy. It moves diagonally toward the nearest tile to the player, covering one or more tiles in an odd turn. It also attacks diagonally. When it attacks, it reduces the player's steps by 2.



Charger

Charger is a multi-tile mover enemy. It moves straight toward the nearest tile to the player, covering one or more tiles in an even turn. It attacks in the four cardinal directions(up, down, left, or right) and attacks the player if within range. When it attacks, it reduces the player's steps by 2.

Game Characters - continued



Archer

Archer is the long-range enemy. It does not move but instead prepares to shoot on the first turn, and on the following turn, it fires an arrow at the player's previous tile. This behavior forces the player to keep moving and avoid staying on the same tile. When it hits the player, it reduces the player's step by 2.



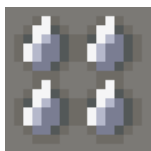
Cannon

Cannon is a long-range enemy. It does not move but begins charging on the first turn. After two turns, it fires a cannonball if the player is within its range. Cannon attacks straight in a line. When the cannonball hits the player, it reduces the player's step by 3.



King

King is the final boss of the game. It does not move and is fixed on the top of the board. Each time the player attacks the King, it summons one of the enemies listed above. Also, it sends the player back to the starting point.



Trap

Traps do not directly attack enemies. Instead, they have a 2-turn delay before spikes emerge. If the player is standing on the trap when it activates, they will be damaged.



Merchant

The merchant is a non-playable character who appears in the store. It is wearing a green robe with a huge backpack on its back.

Game Resources

Items

Items are used either after defeating an enemy or when the player reaches the store.

Chocolate



Chocolate is a basic healing item that has an 80% chance to drop from enemies. When consumed, it restores 3 steps to the player.

Omurice



Omurice is a rare healing item that has a 15% chance to drop from enemies. When consumed, it restores 5 steps to the player.

Pizza



Pizza is a unique healing item that has a 5% chance to drop from enemies. When consumed, it restores all steps to the player.

AXE



An Axe is a weapon that players can purchase from the store.

It comes in four rarity levels: Common, Rare, Unique, and Legendary.

Each rarity affects the weapon's durability, price, and appearance

- Common: Durability 1, Price 3 steps, 55% probability
- Rare: Durability 2, Price 4 steps, 30% probability

- Unique: Durability 3, Price 5 steps, 10% probability
- Legendary: Durability 5, Price 6, 5% probability

Game Resources - continued

All weapons are equipped immediately upon acquisition. If the player picks up a new weapon while already holding one, it will automatically be replaced only if the new weapon has higher durability. The Axe attacks all adjacent tiles around the player (1-tile radius), making it effective for close-range combat.

Spear



A Spear is a weapon that players can purchase from the store.

It comes in four rarity levels: Common, Rare, Unique, and Legendary.

Each rarity affects the weapon's durability, price, and appearance

- Common: Durability 1, Price 1 steps, 55% probability
- Rare: Durability 2, Price 2 steps, 30% probability
- Unique: Durability 3, Price 3 steps, 10% probability
- Legendary: Durability 5, Price 4, 5% probability

The spear attacks a single enemy located two tiles behind the player.

Shield



A Shield is armor that players can purchase from the store.

It comes in four rarity levels: Common, Rare, Unique, and Legendary.

Each rarity affects the weapon's durability, price, and appearance

- Common: Durability 1, Price 1 steps, 55% probability
- Rare: Durability 2, Price 2 steps, 30% probability
- Unique: Durability 3, Price 3 steps, 10% probability
- Legendary: Durability 5, Price 4, 5% probability

The shield blocks incoming attacks by consuming its durability whenever the player is attacked.

Game Resources - continued

Boots



Boots are armor that players can purchase from the store

It comes in four rarity levels: Common, Rare, Unique, and Legendary.

Each rarity affects the weapon's durability, price, and appearance

- Common: Durability 1, Price 1 steps, 55% probability
- Rare: Durability 2, Price 2 steps, 30% probability
- Unique: Durability 3, Price 3 steps, 10% probability
- Legendary: Durability 5, Price 4, 5% probability

The boots block the trap's attack by consuming durability whenever the player steps on the activated trap.

Game Environment

Stage

This game contains 3 stages, and it has its own themed environment.

Stage Themes

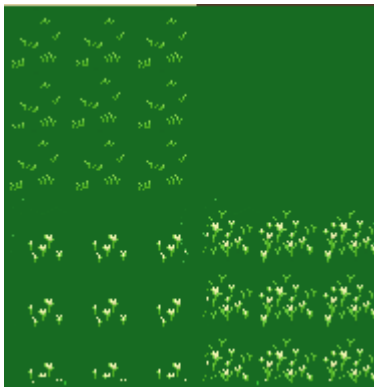
- Chessboard
- Forest
- Castle

Chess board



This stage features a chessboard theme, designed to let players know that the game is inspired by chess.

Forest



This forest-themed stage represents the start of the player's journey, reflecting the common RPG trope of beginning in a forest dungeon.

Castle



This stage features a castle-themed, representing the royal residence. As the Chess King is the final boss, the last battle takes place within his own castle.

Game Controls

Keyboard

The keyboard will be used for characters. All keyboard bindings are mappable in the controls menu.

Default Setting

- **Move Up:** W
- **Move Down:** S
- **Move Left:** A
- **Move Right:** D
- **Restart:** R
- **Leave:** ESC

Visual Design

Tiles

To maintain a sense of continuity in the player's journey, the game's themes are designed to transition naturally without abrupt shifts.

Characters

Sprites and their animations will consist of pixel-style graphics.

Animations

The following lists the type of animations needed for each character.

Player

- Idle
- Walk
- Attack
- Attacked
- Defeated

Pawn

- Idle
- Attack
- Attackable
- Attacked
- Defeated

Merchant

- Idle

Cannon

- Idle horizontal
- Idle vertical
- Attack horizontal
- Attack vertical
- Attackable horizontal
- Attackable vertical
- Defeated horizontal
- Defeated vertical

Visual Design – continued

Bishop

- Idle
- Attack
- Attackable
- Attacked
- Defeated

Archer

- Idle
- Attack
- Attackable
- Attacked
- Defeated

Arrow

- Arrow

Tile

- Trap On
- Trap Off
- Attackable

Audio Design

Music

All the music is well composed and fits each stage perfectly.

Stage 1

Stage 1 features rocking drumbeats that set the tone for the start of the adventure.

Stage 2

Stage 2 features the music taking on a more casual tone. This stage represents the start of the player's journey.

Stage 3, Boss

In the castle stage, the music grows more tense, it becomes even faster and more dramatic to match the climax of the game.

Store

In the shop level, where players can move around without pressure, the music is calmer to create a relaxed atmosphere

Sound Effects

The game's sound effects also align well with the overall concept.

- Turn change
- Moving
- Attack
- Item buying
- Cannon
- Arrow

Behavior Design

Enemies

When the player moves from tile to tile, each enemy calculates the closest tile it can reach. Afterward, the enemy checks whether it can attack the player. Based on this, the enemy chooses to either attack or move.